

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 & Machine Learning Algorithms Laboratory		
Academic year	2025-2026 Even	Batch	2023-2027
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Experiment 4: Binary Classification using Linear and Kernel-Based Models

1.Aim and Objective:

The aim of this experiment is to classify emails as spam or ham using Logistic Regression and Support Vector Machine classifiers and to analyze the impact of hyperparameter tuning on the performance of linear and kernel-based models. The objectives are to load and preprocess the Spambase dataset by handling missing values and standardizing features, perform exploratory data analysis to understand class distribution and feature relationships, implement baseline Logistic Regression and SVM models, apply hyperparameter tuning using Grid Search and/or Randomized Search, and evaluate the models using accuracy, precision, recall, F1 score, and 5-fold cross-validation.

2. Dataset Description:

The Spambase dataset contains numerical features extracted from email content along with a binary class label indicating whether an email is spam or ham. It consists of 4,601 samples with 57 input features and one target variable. The class label is 1 for spam and 0 for non-spam emails. Since all features are numerical, the dataset is suitable for applying classification algorithms after preprocessing and feature scaling.

3. Preprocessing Steps:

- Loaded the dataset using Pandas.
- Checked for missing values and handled them using mean imputation (if any were present).
- Separated input features and the target variable.

- Applied standardization to scale all numerical features using StandardScaler so that they have zero mean and unit variance.
- Verified the processed data before proceeding to model training.

4. Implementation Details:

- The Spambase dataset was loaded using Pandas and examined for missing values and data consistency.
- Numerical features were standardized using StandardScaler to improve the performance of Logistic Regression and SVM classifiers.
- Exploratory Data Analysis (EDA) was carried out to study class distribution and feature correlations.
- The dataset was divided into training and testing sets using stratified sampling.
- A baseline Logistic Regression model was trained first, followed by hyperparameter tuning through Grid Search.
- Support Vector Machine models with multiple kernels were implemented, tuned for optimal parameters, and evaluated using performance metrics and 5-fold cross-validation.

5. Visualizations:

5.1. Class Distribution

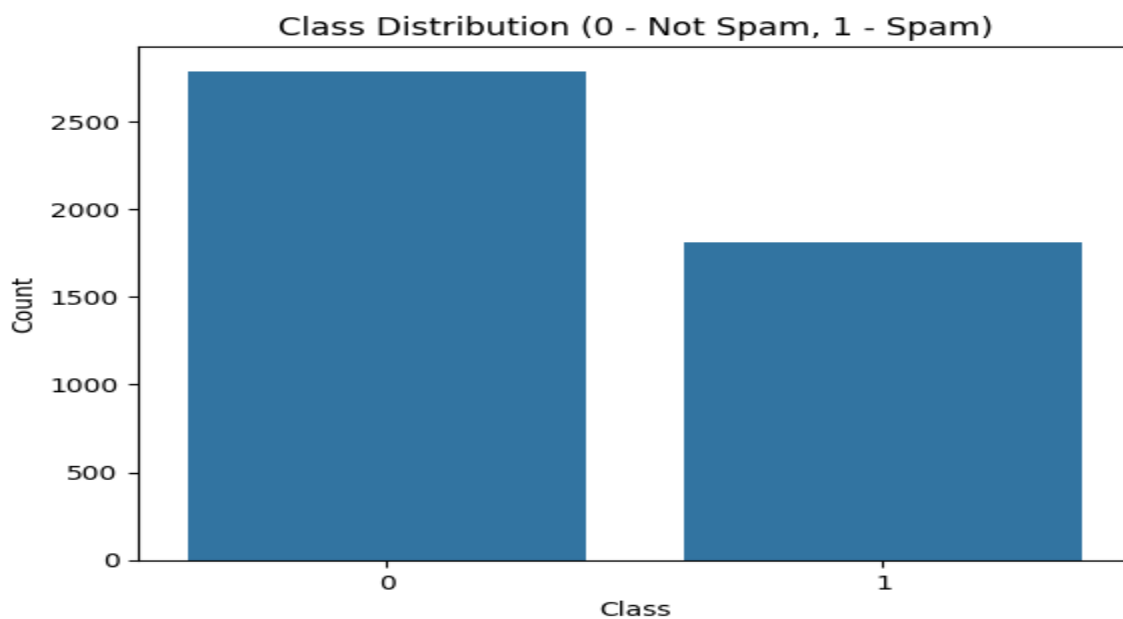


Figure 1: Class distribution of Spam and Non-Spam Emails

5.2. Feature Distribution

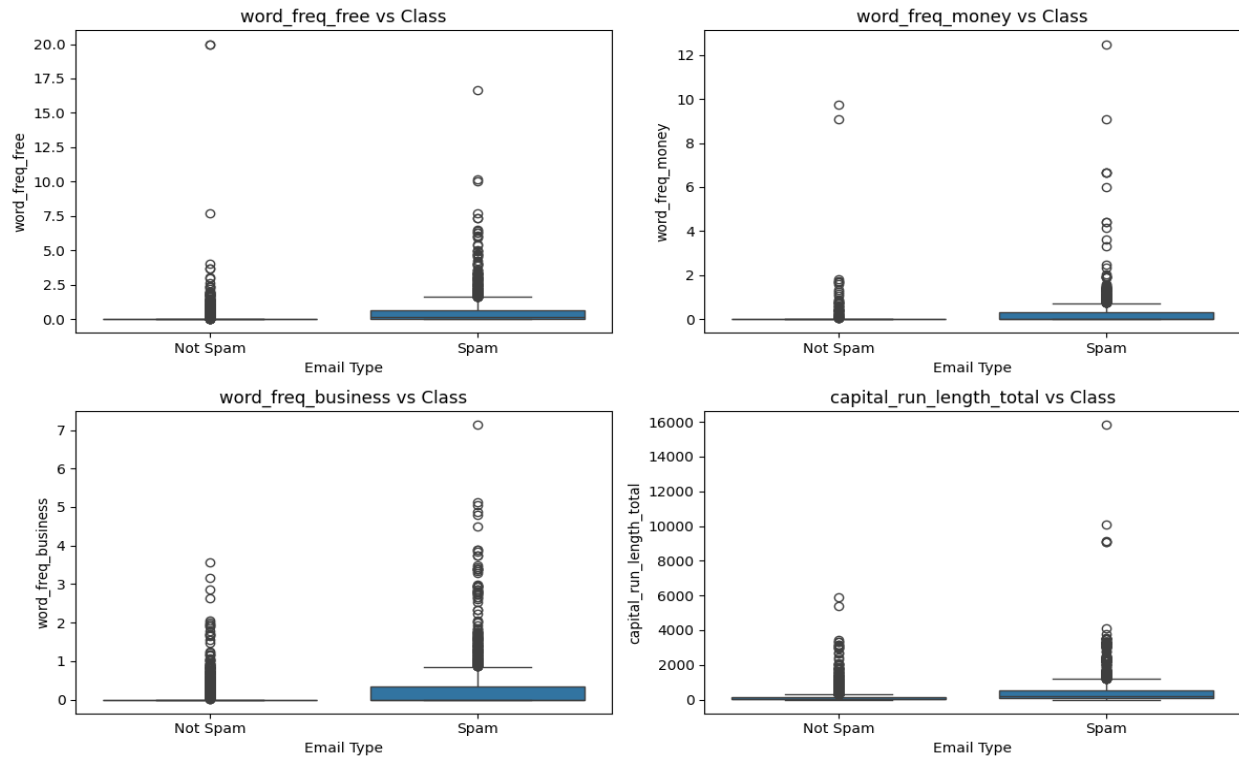


Figure 2: Boxplots for important features

5.3. Logistic regression results

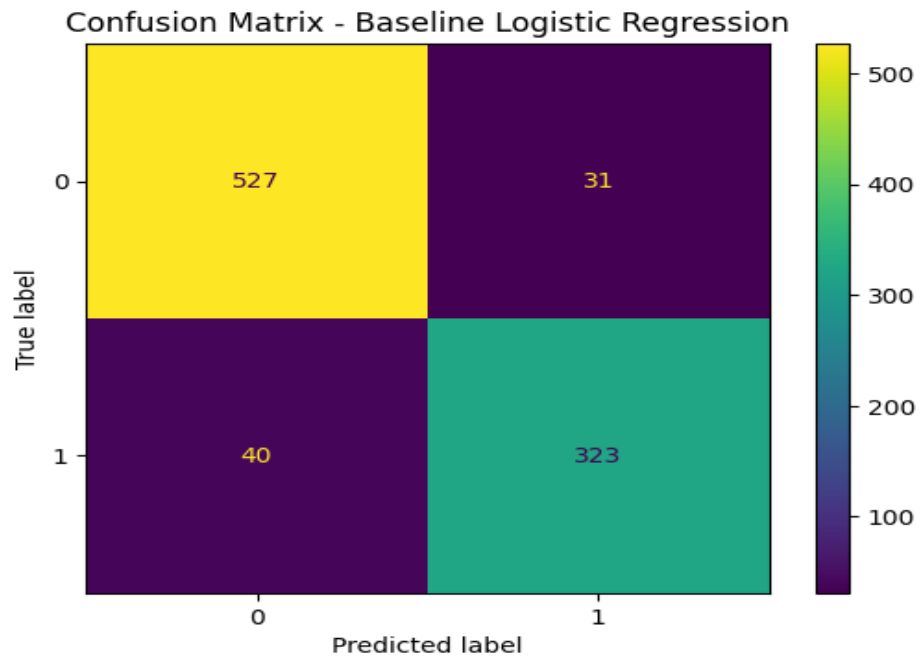


Figure 3: Confusion matrix for Linear regression

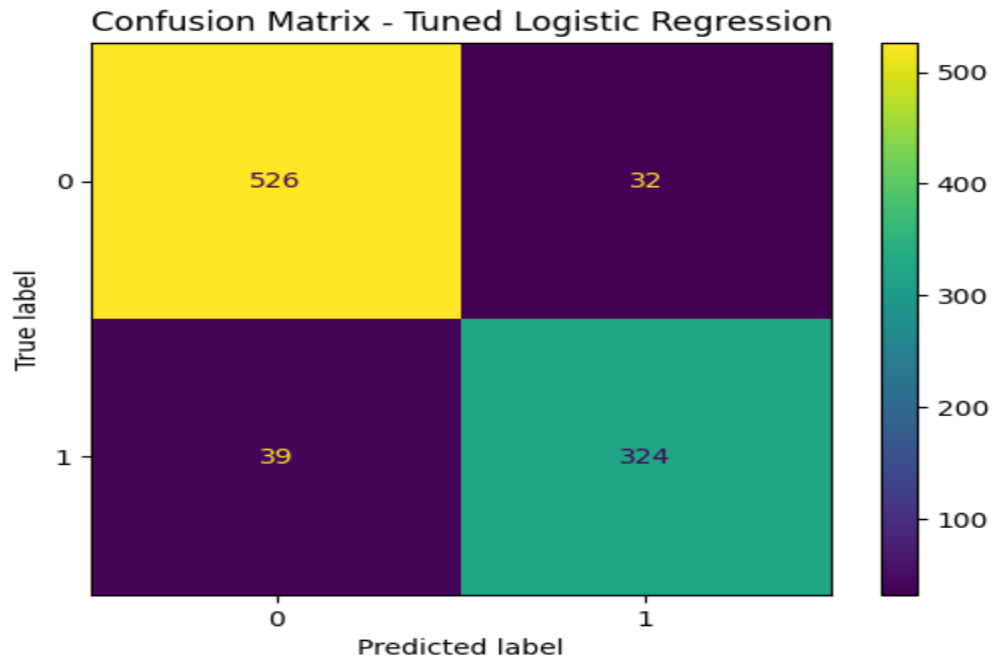


Figure 4: Confusion matrix for Tuned Linear regression

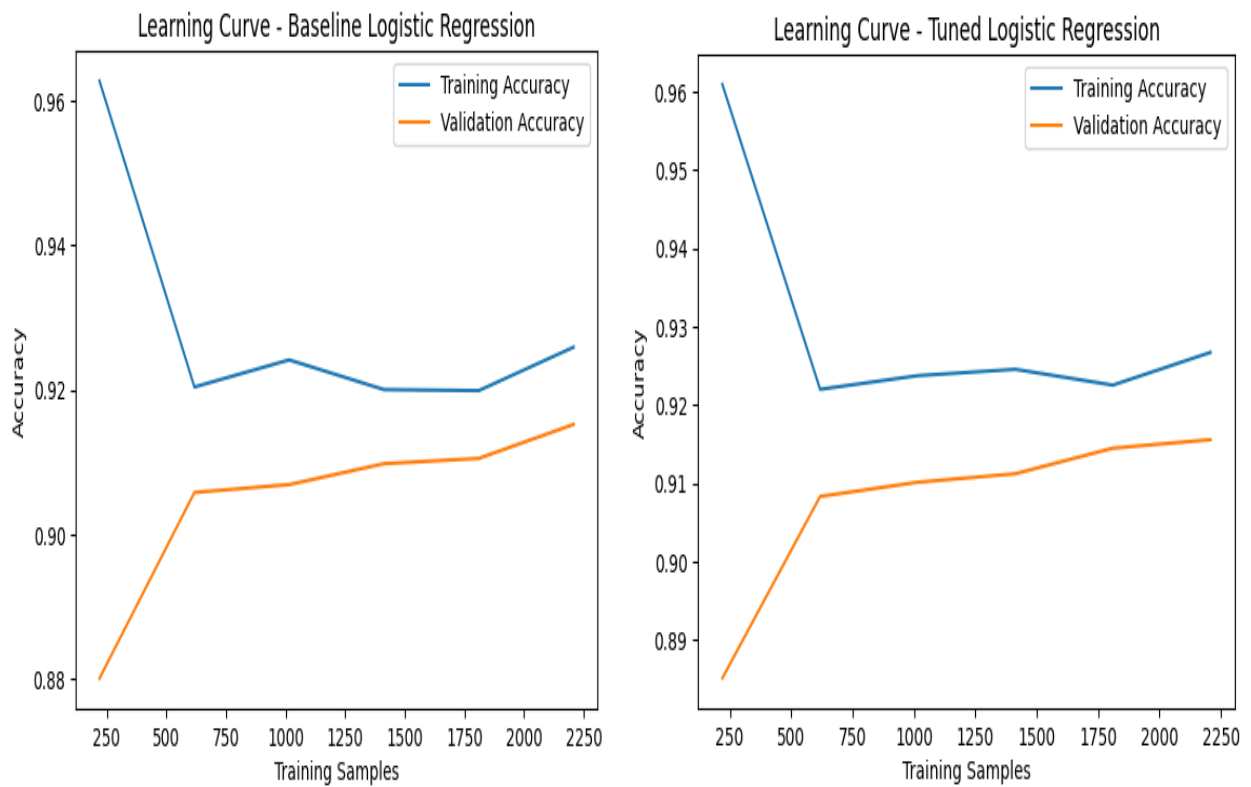


Figure 5&6: Learning curve for Baseline Logistic regression and Tuned Linear regression

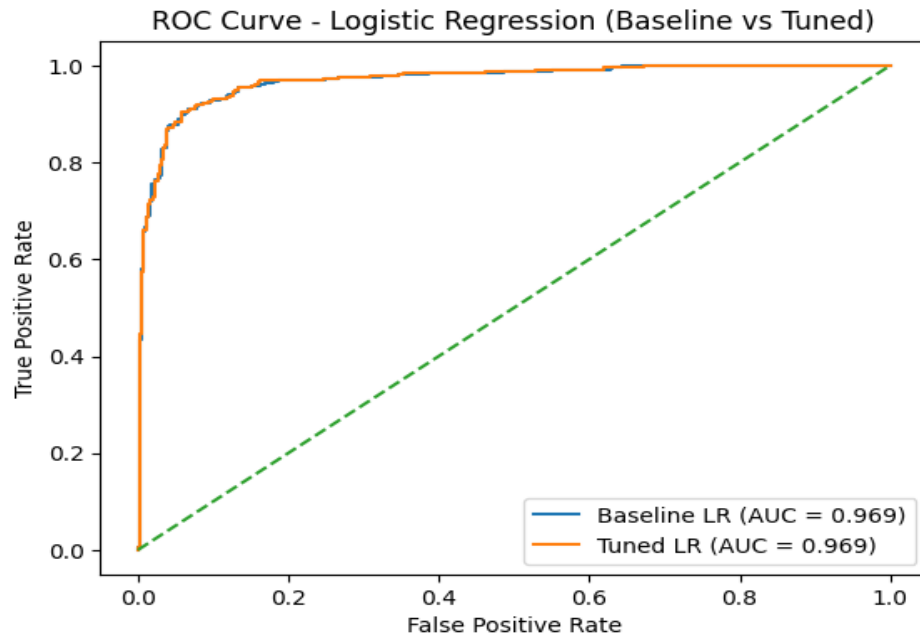


Figure 7: ROC Curve – Logistic Regression (Baseline vs Tuned)

5.4. Support Vector Machine results

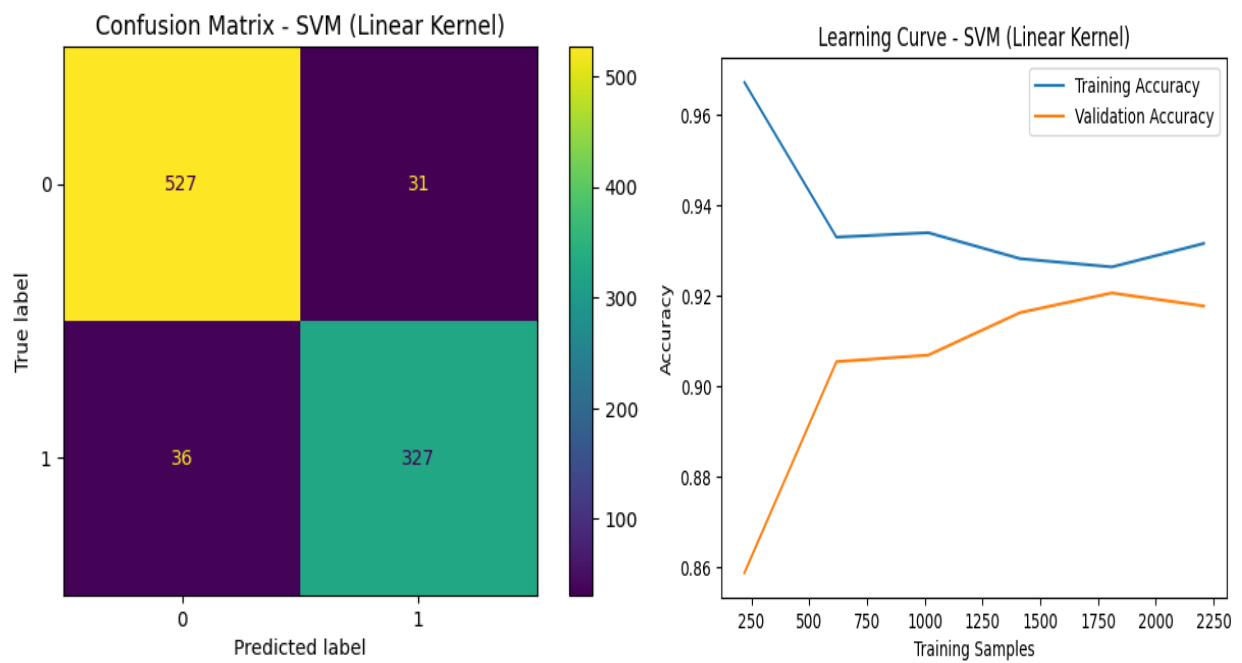


Figure 8&9: Confusion Matrix and Learning Curve of SVM(Linear)

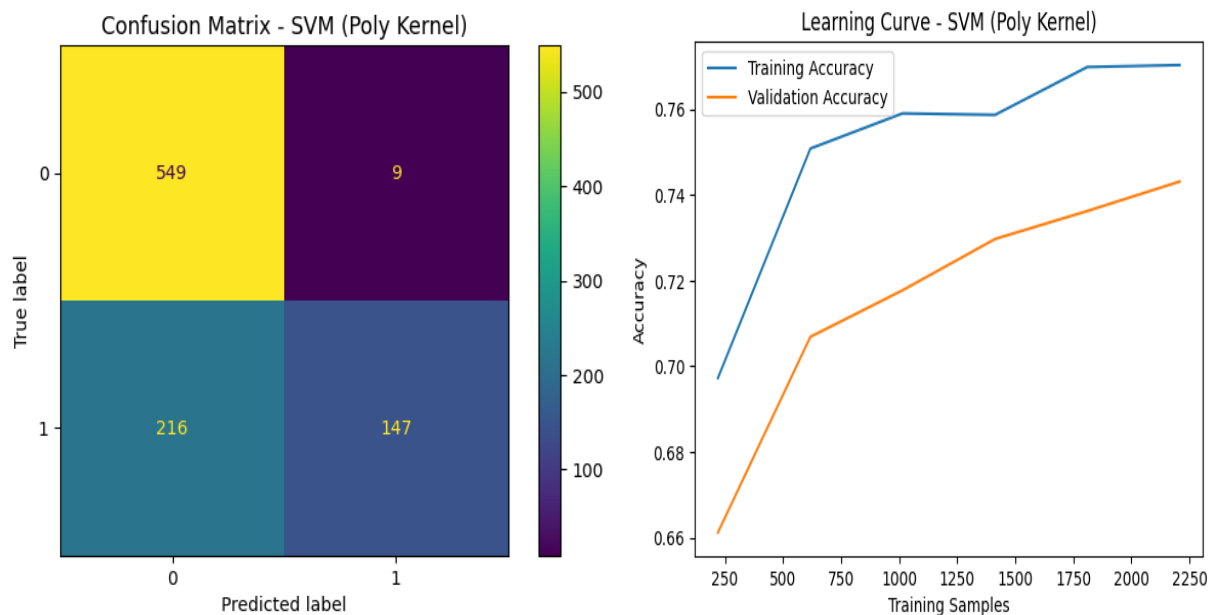


Figure 10&11: Confusion Matrix and Learning Curve of SVM(Poly)

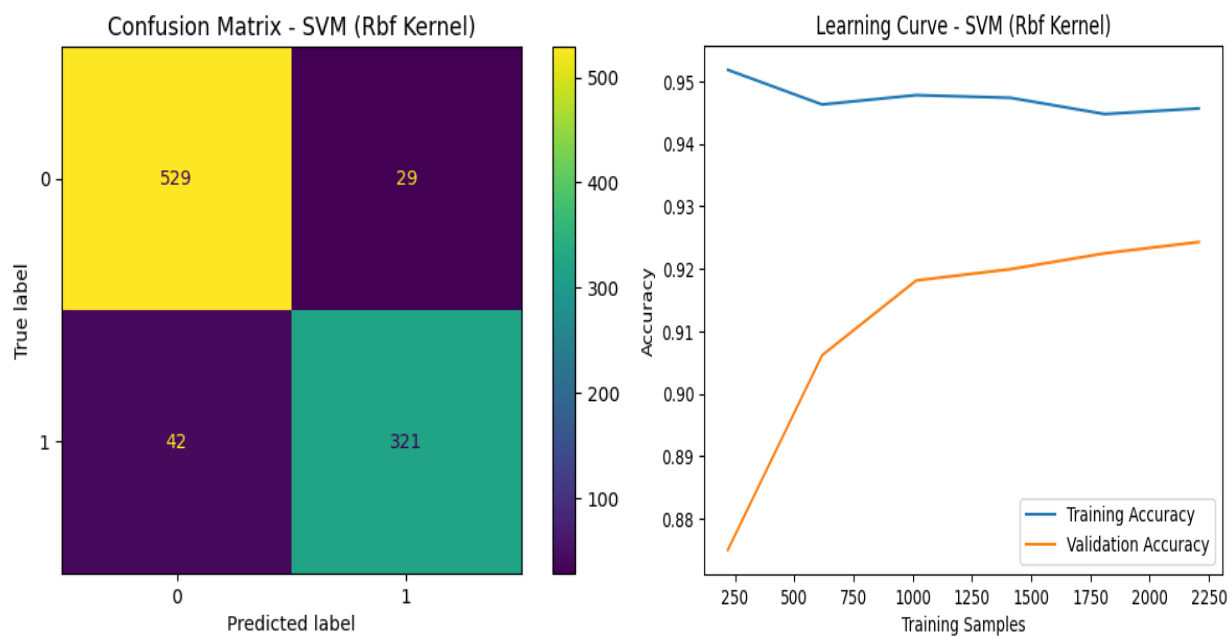


Figure 12&13: Confusion Matrix and Learning Curve of SVM(RBF)

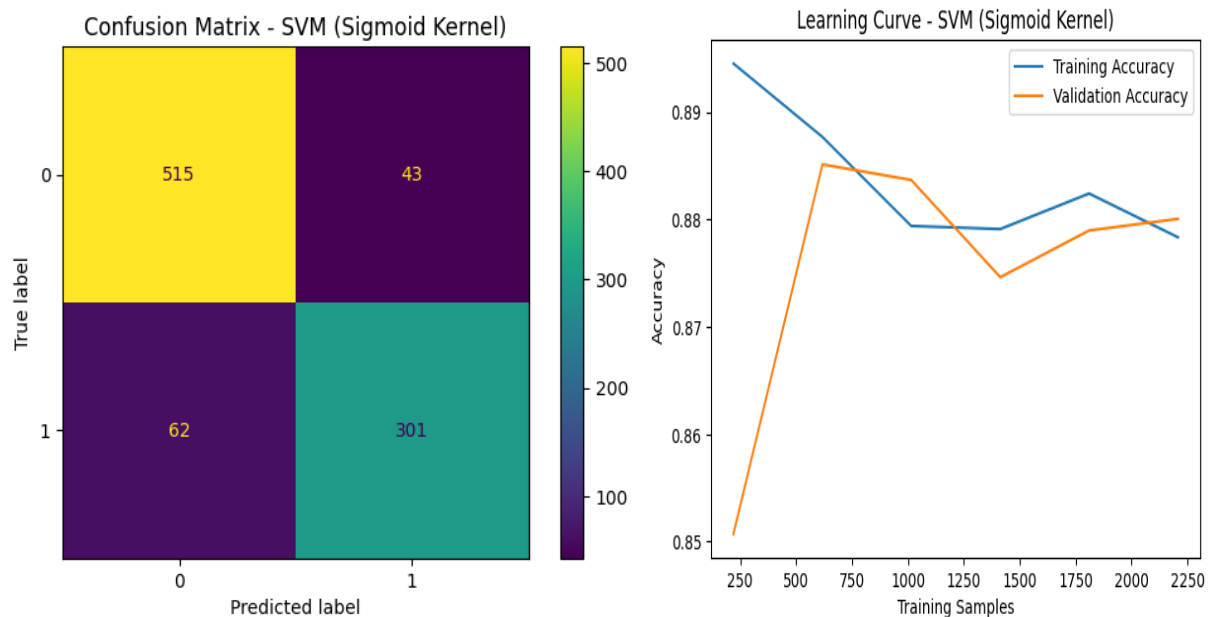


Figure 14&15: Confusion Matrix and Learning Curve of SVM(Sigmoid)

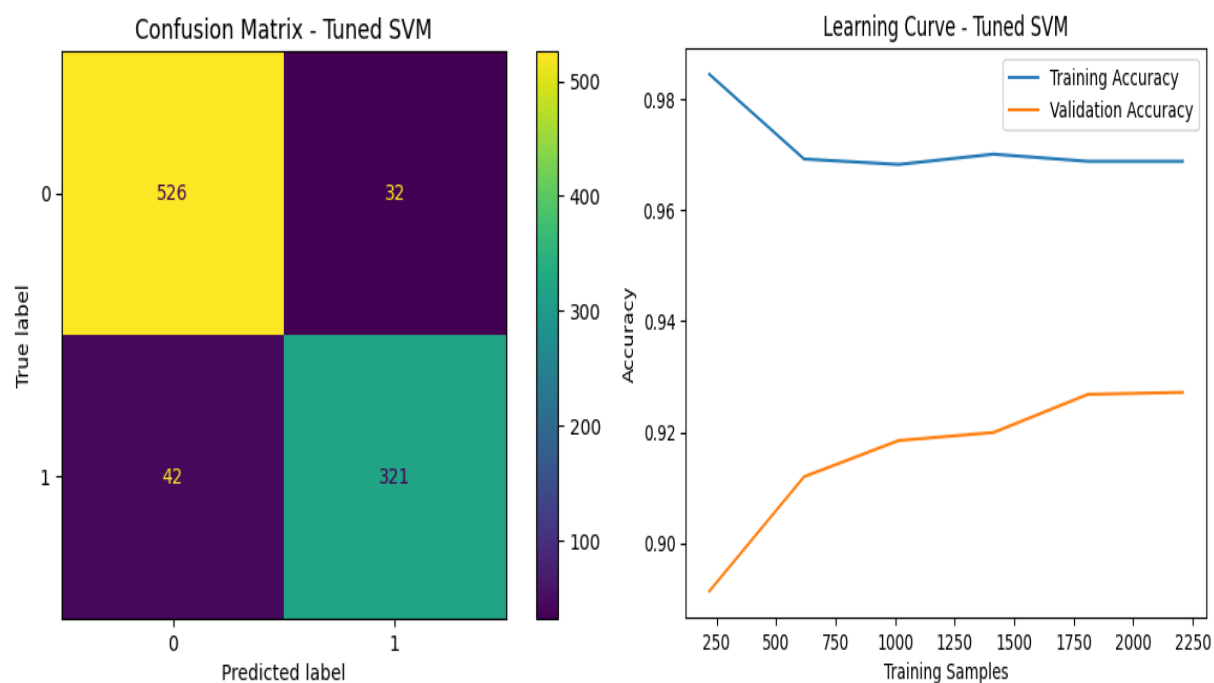


Figure 16&17: Confusion Matrix and Learning Curve of SVM(Tuned)

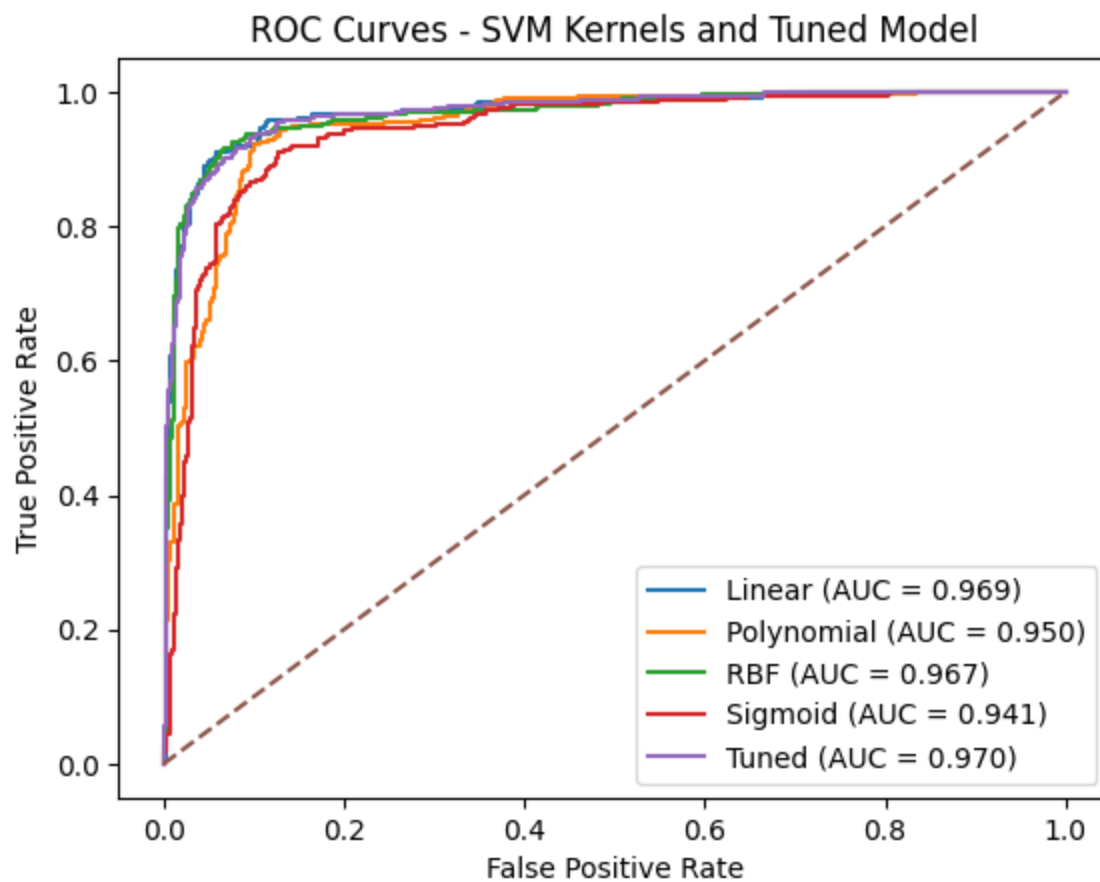


Figure 18: ROC for SVM kernels

6. Performance Tables:

Hyperparameter Tuning Results

Model	Search Method	Best Parameters	Best CV Accuracy
Logistic Regression	Grid Search	penalty = L1, C = 1, solver = Liblinear	0.9155
SVM	Grid Search	kernel = RBF, C = 10, gamma = scale	0.9271

Table 1: Hyperparameter Tuning Results

Logistic Regression Performance

Metric	Value
Accuracy	0.9229
Precision	0.9124
Recall	0.8898
F1 Score	0.9009
Training Time (s)	0.0259

Table 2: Logistic Regression Performance

SVM Kernel-wise Performance

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.9275	0.9070	0.2117
Polynomial	0.7557	0.5664	0.1409
RBF	0.9229	0.9004	0.0891
Sigmoid	0.8859	0.8514	0.0999

Table 3: SVM Kernel-wise Performance

K-Fold Cross-Validation Results (K = 5)

Fold	Logistic Regression	SVM
Fold 1	0.9166	0.9293
Fold 2	0.9076	0.9257
Fold 3	0.9039	0.9166
Fold 4	0.9293	0.9384
Fold 5	0.9202	0.9257
Average	0.9155	0.9271

Table 4: K-Fold Cross-Validation Results (K = 5) for LR and SVM

Comparative Analysis Summary

Criterion	Logistic Regression	SVM
Accuracy	High (0.9155)	Very High (0.9271)
Model Complexity	Low	High
Training Time	Low	High
Interpretability	High	Low

Table 5: Comparative Analysis of Logistic regression and SVM

7. Observations and Conclusion:

The results show that both Logistic Regression and SVM models benefited significantly from feature scaling and hyperparameter tuning. Logistic Regression with L2 regularization provided stable performance with low training time and good interpretability. Among SVM kernels, the RBF kernel achieved the highest accuracy and F1 score, indicating its ability to model complex patterns in the data.

Cross-validation confirmed that the tuned SVM generalized better across different splits. Overall, the experiment concludes that tuned SVM is the best-performing classifier for the Spambase dataset, while Logistic Regression serves as an efficient and interpretable baseline.

Learning Outcomes:

- Understood the working principles of Logistic Regression and Support Vector Machine classifiers for binary classification tasks.
- Learned to preprocess the dataset using feature scaling and missing-value handling to improve both LR and SVM performance.
- Applied hyperparameter tuning techniques such as Grid Search for Logistic Regression and SVM models.
- Evaluated LR and SVM classifiers using accuracy, precision, recall, F1 score, ROC curves, and confusion matrices.
- Compared linear and kernel-based models in terms of performance, complexity, and bias–variance behavior.